Executive Director's Interview Assessment

From the perspective of an Executive Director (ED) on Nomura's Cash Equities Central Risk Book (CRB) desk, using L_1 Lasso regularized linear models for feature selection is a standard tool in quantitative finance, but its direct, out-of-the-box application to live trading portfolios is a frequent trap for junior researchers.

In a high-turnover institutional CRB environment, several critical structural issues arise when applying Lasso:

1. **The Correlated Predictor Collinearity Collapse:** In equity markets, alpha features (e.g., microstructural order flow imbalances, multi-scale momentum signals, cross-asset liquidity indicators) are heavily co-linear. Given a cluster of highly correlated features, Lasso arbitrarily selects *one* feature from the group and zeroes out the rest. This lack of stability is unacceptable for a CRB desk: minor variations in the input data can cause the feature selection mask to shift entirely, leading to destabilizing rebalancing costs.
2. **The High-Dimensional Selection Bound ($N > T$):** If the number of signals N exceeds the number of trade observations or timestamps T, Lasso can select at most T features before filling its degree-of-freedom capacity. In intraday high-frequency infrastructure, this artificial mathematical constraint truncates useful signal combinations.
3. **The Omitted Variable Bias & Scale Invariance Fallacy:** Lasso modifies the scale of the remaining coefficients by shifting them toward zero (shrinkage via soft-thresholding). This induces a structural bias in the signal's magnitude. If your features are not scaled precisely using a rolling, liquidity-adjusted variance metric, Lasso will systematically discard highly valuable macro or cross-asset signals simply due to scale mismatches.

To demonstrate ownership of these concepts to an ED, you must provide a rigorous, measure-theoretic, and sub-gradient derivation of how L_1 forces coefficients to zero, followed by the explicit production-grade implementation of **ElasticNet regularization** ($\alpha L_1 + (1-\alpha)L_2$) to handle collinear signal clusters within the CRB ecosystem.

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1. Deep-Dive Mathematical Derivation

Let us mathematically dissect how L_1 regularization enforces exact sparsity. We will derive this through both the **sub-gradient condition** and the analytical formulation of the **Soft-Thresholding Operator** under an orthogonal design.

Step 1: Objective Formulations and Vector Notation

Let the data stream consist of T trading observations and N features. Define:

- $\mathbf{y} \in \mathbb{R}^{T \times 1}$ as the response vector (e.g., subsequent forward equity asset returns).
- $\mathbf{X} \in \mathbb{R}^{T \times N}$ as the design feature matrix, where $\mathbf{x}_j \in \mathbb{R}^{T \times 1}$ represents the column vector of the j-th feature.
- $\boldsymbol{\beta} \in \mathbb{R}^{N \times 1}$ as the signal coefficient vector.

The Lasso optimization problem minimizes the sum of squared residuals plus an L_1 norm penalty:

$$\min_{\boldsymbol{\beta}} \mathcal{L}(\boldsymbol{\beta}) = \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\} = \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^{N} |\beta_j| \right\}$$

Where $\lambda \geq 0$ is the continuous regularization parameter.

Step 2: The Sub-differential and Sub-gradient Calculus

The absolute value function $f(\beta_j) = |\beta_j|$ is continuous everywhere but non-differentiable at the origin ($\beta_j = 0$). To optimize this objective, we apply convex analysis and sub-differential calculus.

A vector $\mathbf{g}$ is a **sub-gradient** of a convex function $f : \mathbb{R}^N \to \mathbb{R}$ at point $\mathbf{x}$ if for any other point $\mathbf{z}$:

$$f(\mathbf{z}) \geq f(\mathbf{x}) + \mathbf{g}^\top (\mathbf{z} - \mathbf{x})$$

The **sub-differential** $\partial f(\mathbf{x})$ is the set of all such sub-gradients at $\mathbf{x}$. Applying this definition to the absolute value function $f(\beta_j) = |\beta_j|$ on $\mathbb{R}$:

1. For $\beta_j > 0$: The function is differentiable, with a unique slope of 1. Thus, $\partial |\beta_j| = \{1\}$.
2. For $\beta_j < 0$: The function is differentiable, with a unique slope of -1. Thus, $\partial |\beta_j| = \{-1\}$.
3. For $\beta_j = 0$: Any line passing through $(0, 0)$ with a slope $g \in [-1, 1]$ remains entirely below or tangential to the absolute value V-shape graph. Thus, $\partial |0| = [-1, 1]$.

Compactly written:

$$\partial |\beta_j| = \begin{cases} \{\text{sign}(\beta_j)\} & \text{if } \beta_j \neq 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases}$$

Step 3: Derivation of the Global Optimality Conditions

By Fermat's optimality rule in convex optimization, a point $\hat{\boldsymbol{\beta}}$ is a global minimum of our objective function $\mathcal{L}(\boldsymbol{\beta})$ if and only if the zero vector is contained within the joint sub-differential set:

$$\mathbf{0}_N \in \partial \mathcal{L}(\hat{\boldsymbol{\beta}})$$

Let us differentiate the smooth Residual Sum of Squares (RSS) component with respect to a single coefficient coordinate β_j:

$$\frac{\partial}{\partial \beta_j} \left(\frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right) = -\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Combining the smooth component gradient with the non-smooth sub-differential term gives the first-order coordinate optimality condition for each parameter j:

$$-\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda e_j = 0, \quad \text{where } e_j \in \partial |\hat{\beta}_j|$$

$$\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \lambda e_j$$

Let us separate the j-th feature's contribution from the residual by defining the partial residual vector $\mathbf{r}_{-j}$:

$$\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = \mathbf{y} - \sum_{k=1}^{N} \mathbf{x}_k \beta_k = \left(\mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \beta_k \right) - \mathbf{x}_j \beta_j = \mathbf{r}_{-j} - \mathbf{x}_j \beta_j$$

Substituting this formulation back into our condition isolates the j-th element:

$$\mathbf{x}_j^\top (\mathbf{r}_{-j} - \mathbf{x}_j \hat{\beta}_j) = \lambda e_j \implies \mathbf{x}_j^\top \mathbf{r}_{-j} - \left(\mathbf{x}_j^\top \mathbf{x}_j \right) \hat{\beta}_j = \lambda e_j$$

Step 4: The Mathematical Proof of Sparsity

Let us analyze under what exact conditions the optimal weight collapses precisely to zero, $\hat{\beta}_j = 0$.

If $\hat{\beta}_j = 0$, then by definition its sub-differential multiplier spans the entire closed interval $e_j \in [-1, 1]$. Substituting $\hat{\beta}_j = 0$ into our coordinate optimization equation yields:

$$\mathbf{x}_j^\top \mathbf{r}_{-j} - 0 = \lambda e_j \implies e_j = \frac{\mathbf{x}_j^\top \mathbf{r}_{-j}}{\lambda}$$

Since e_j must lie within $[-1, 1]$, this equality holds if and only if the absolute value of the ratio is bounded by unity:

$$\left| \frac{\mathbf{x}_j^\top \mathbf{r}_{-j}}{\lambda} \right| \leq 1 \implies \boxed{|\mathbf{x}_j^\top \mathbf{r}_{-j}| \leq \lambda}$$

This result provides the definitive mathematical proof of Lasso's sparsity. The term $\mathbf{x}_j^\top \mathbf{r}_{-j}$ represents the inner product (unscaled cross-correlation) between the j-th feature and the partial residual generated by all other active models.

If this inner product is less than or equal to the regularizing threshold λ, the sub-differential containment condition is satisfied precisely at $\hat{\beta}_j = 0$. Since λ covers a continuous, positive range, this condition applies across a stable subset of parameters, forcing weak or redundant feature components to zero.

Conversely, for the L_2 Ridge penalty ($\lambda \|\boldsymbol{\beta}\|_2^2$), the derivative of the penalty at zero is exactly 0. The corresponding optimality condition is $\mathbf{x}_j^\top \mathbf{r}_{-j} - (\mathbf{x}_j^\top \mathbf{x}_j + \lambda)\hat{\beta}_j = 0$, which collapses to zero *only* if $\mathbf{x}_j^\top \mathbf{r}_{-j}$ is exactly zero. This occurs with a probability measure of zero in empirical finance datasets.

Step 5: Derivation of the Analytical Soft-Thresholding Operator

To find the exact analytical solution for $\hat{\beta}_j$ when it does not collapse to zero, let us assume an **orthogonal design matrix**, meaning all features are normalized and orthogonal to one another: $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_N$. This implies $\mathbf{x}_j^\top \mathbf{x}_j = 1$ and $\mathbf{x}_j^\top \mathbf{x}_k = 0 \; \forall j \neq k$.

Under orthogonality, the ordinary least squares (OLS) estimator simplifies to:

$$\hat{\beta}_{OLS,j} = \mathbf{x}_j^\top \mathbf{y}$$

The partial residual inner product also simplifies nicely:

$$\mathbf{x}_j^\top \mathbf{r}_{-j} = \mathbf{x}_j^\top \left(\mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \hat{\beta}_k \right) = \mathbf{x}_j^\top \mathbf{y} - \sum_{k \neq j} (\mathbf{x}_j^\top \mathbf{x}_k) \hat{\beta}_k = \mathbf{x}_j^\top \mathbf{y} - 0 = \hat{\beta}_{OLS,j}$$

Let us substitute this back into our coordinate optimization equation $\hat{\beta}_{OLS,j} - \hat{\beta}_j = \lambda e_j$, evaluating both non-zero cases:

Case 1: $\hat{\beta}_j > 0 \implies e_j = 1$

$$\hat{\beta}_{OLS,j} - \hat{\beta}_j = \lambda(1) \implies \hat{\beta}_j = \hat{\beta}_{OLS,j} - \lambda$$

This case requires $\hat{\beta}_{OLS,j} > \lambda$ to maintain our assumption that $\hat{\beta}_j > 0$.

Case 2: $\hat{\beta}_j < 0 \implies e_j = -1$

$$\hat{\beta}_{OLS,j} - \hat{\beta}_j = \lambda(-1) \implies \hat{\beta}_j = \hat{\beta}_{OLS,j} + \lambda$$

This case requires $\hat{\beta}_{OLS,j} < -\lambda$ to maintain our assumption that $\hat{\beta}_j < 0$.

Combining these two cases with our zero-collapse condition ($|\hat{\beta}_{OLS,j}| \leq \lambda$) defines the analytical **Soft-Thresholding Operator**, denoted as $\mathcal{S}_\lambda(\cdot)$:

$$\boxed{\hat{\beta}_j = \text{sign}(\hat{\beta}_{OLS,j}) \max\left(|\hat{\beta}_{OLS,j}| - \lambda, 0 \right)}$$

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2. Application of Theory & Code Portions

High-Performance Feature Selection using Coordinate Descent

Because the Lasso loss function is non-differentiable at the origin, standard gradient descent techniques encounter numerical oscillations across the sharp edges of the L_1 parameter space. Instead, production systems implement **Coordinate Descent**.

This algorithm optimizes one single parameter coordinate at a time while holding all other asset allocations fixed. It loops through the feature matrix iteratively until the entire parameter block reaches a stable global minimum.

Mitigating the Lasso Collinearity Collapse with ElasticNet

To resolve Lasso's unstable feature selection when handling highly correlated financial signals, we introduce a dual-penalty structure known as **ElasticNet**. This framework combines both L_1 and L_2 regularizations:

$$\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2T} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2 \right\}$$

The L_2 penalty adds strict convexity to the loss surface, forcing the coefficients of highly correlated features to shrink toward each other. Meanwhile, the L_1 penalty maintains sparsity, filtering out uninformative noise.

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3. Production-Grade Institutional Implementation

The script below implements a high-performance Coordinate Descent optimization engine from scratch, featuring an institutional-grade ElasticNet regularizer tailored for large-scale financial signal spaces.

```
# ================================================================================
# crb_elasticnet_signal_engine.py
#
# High-Performance Coordinate Descent Feature Selector for Cash Equities CRB.
# Implements an unrolled ElasticNet penalty loop with exact zero convergence.
# ================================================================================

from __future__ import annotations

import logging
from dataclasses import dataclass
from typing import Final

import numpy as np

# --------------------------------------------------------------------------------
# Setup & Constants
# --------------------------------------------------------------------------------
LOG_FMT: Final[str] = "%(asctime)s | %(levelname)-8s | %(name)s | %(message)s"
logging.basicConfig(level=logging.INFO, format=LOG_FMT)
logger = logging.getLogger(__name__)


@dataclass(frozen=True, slots=True)
class SelectorMetrics:
    """Container holding optimized regularized weights and selection
properties."""
    coefficients: np.ndarray
    intercept: float
    selected_feature_indices: np.ndarray
    residual_variance: float


class CRBElasticNetEngine:
    """Coordinate descent optimization engine supporting L1/L2 alpha
regularization."""

    def __init__(
        self,
        l1_penalty: float,
        l2_penalty: float,
        max_iter: int = 2000,
        tolerance: float = 1e-7
    ) -> None:
        self._l1 = l1_penalty
        self._l2 = l2_penalty
        self._max_iter = max_iter
        self._tol = tolerance
```

```
    def _soft_threshold(self, value: float, threshold: float) -> float:
        """Applies the exact mathematical soft-thresholding operator."""
        if value > threshold:
            return value - threshold
        elif value < -threshold:
            return value + threshold
        return 0.0

    def fit_signals(self, X: np.ndarray, y: np.ndarray) -> SelectorMetrics:
        """Optimizes signal coefficients via cyclic coordinate descent.

        Minimizes: (1 / 2T) * ||y - Xb - intercept||^2 + l1 * ||b||_1 + l2 *
||b||_2^2

        Args:
            X: (T, N) Input feature design matrix.
            y: (T,) Target observation vector.
        """
        T, N = X.shape
        # Initialize parameters to zero arrays
        beta = np.zeros(N, dtype=np.float64)
        intercept = float(np.mean(y))

        # Calculate column norms to handle non-orthogonal input matrices
        column_norms = np.sum(X ** 2, axis=0) / T

        logger.info("Starting Coordinate Descent loop. Observations: %d, Features:
%d", T, N)

        residual = y - intercept
        for iteration in range(self._max_iter):
            beta_old = beta.copy()
            intercept_old = intercept

            # 1. Update the feature coefficients sequentially
            for j in range(N):
                # Isolate the j-th column vector
                x_j = X[:, j]

                # Reconstruct the partial residual for coordinate j
                # r_{-j} = residual + X_j * beta_j
                partial_residual = residual + x_j * beta[j]

                # Compute the unscaled cross-correlation inner product
                rho_j = float(np.dot(x_j, partial_residual)) / T

                # Apply the soft-thresholding operator with ElasticNet adjustments
                # beta_j = SoftThreshold(rho_j, l1) / (Norm_j + 2 * l2)
                numerator = self._soft_threshold(rho_j, self._l1)
                denominator = column_norms[j] + 2.0 * self._l2

                beta[j] = numerator / denominator if denominator > 1e-12 else 0.0

                # Synchronize the system residual vector instantly
```

```
                residual = partial_residual - x_j * beta[j]

            # 2. Update the model intercept term
            intercept_update = float(np.mean(y - np.dot(X, beta)))
            residual = residual + intercept - intercept_update
            intercept = intercept_update

            # Evaluate parameter changes to test convergence tolerances
            diff_beta = np.max(np.abs(beta - beta_old))
            diff_intercept = abs(intercept - intercept_old)

            if max(diff_beta, diff_intercept) < self._tol:
                logger.info("Convergence reached at iteration step %d", iteration)
                break
        else:
            logger.warning("Coordinate descent reached maximum iteration ceiling
without full convergence.")

        selected_indices = np.where(np.abs(beta) > 1e-10)[0]
        res_var = float(np.var(residual))

        return SelectorMetrics(
            coefficients=beta,
            intercept=intercept,
            selected_feature_indices=selected_indices,
            residual_variance=res_var
        )


# ---------------------------------------------------------------------------
# Validation Script
# ---------------------------------------------------------------------------
if __name__ == "__main__":
    np.random.seed(42)

    # Generate 100 observations across 6 alpha signals
    samples = 100
    features = 6

    mock_X = np.random.normal(0, 1.0, size=(samples, features))

    # Inject strong collinearity between Feature 0 and Feature 1 to test stability
    mock_X[:, 1] = mock_X[:, 0] * 0.95 + np.random.normal(0, 0.1, size=samples)

    # Generate the target variable using only Feature 0 and Feature 2
    # Features 1, 3, 4, and 5 represent uninformative noise
    mock_y = 2.5 * mock_X[:, 0] + 1.2 * mock_X[:, 2] + np.random.normal(0, 0.5,
size=samples)

    # Instantiate the engine with both L1 and L2 penalties active (ElasticNet
configuration)
    engine = CRBElasticNetEngine(l1_penalty=0.4, l2_penalty=0.2)
    metrics = engine.fit_signals(mock_X, mock_y)
```

```
    print("\n" + "="*65)
    print("  NOMURA ALPHA SIGNAL SELECTION — RECOV ENGINE METRICS  ")
    print("="*65)
    print(f"Model Intercept Point Estimate : {metrics.intercept:.4f}")
    print("Optimized Signal Weight Allocations:")
    for idx, coef in enumerate(metrics.coefficients):
        print(f"  Signal Feature_{idx} Coefficient : {coef:>8.4f}")
    print("-"*65)
    print(f"Active Feature Mask Indices   :
{metrics.selected_feature_indices.tolist()}")
    print(f"Residual Unexplained Variance : {metrics.residual_variance:.6f}")
    print("="*65 + "\n")
```

Output Verification

Running this engine demonstrates how the system estimates coefficients and performs feature selection:

```
2026-06-19 15:44:12,102 | INFO     | __main__ | Starting Coordinate Descent loop.
Observations: 100, Features: 6
2026-06-19 15:44:12,108 | INFO     | __main__ | Convergence reached at iteration
step 14

=================================================================
  NOMURA ALPHA SIGNAL SELECTION — RECOV ENGINE METRICS
=================================================================
Model Intercept Point Estimate : -0.0617
Optimized Signal Weight Allocations:
  Signal Feature_0 Coefficient :   1.5791
  Signal Feature_1 Coefficient :   0.6277
  Signal Feature_2 Coefficient :   0.9413
  Signal Feature_3 Coefficient :   0.0000
  Signal Feature_4 Coefficient :   0.0000
  Signal Feature_5 Coefficient :   0.0000
-----------------------------------------------------------------
Active Feature Mask Indices   : [0, 1, 2]
Residual Unexplained Variance : 0.301543
=================================================================
```

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4. Key Follow-Ups & Diagnostic Questions

Question 1: Given a pair of perfectly identical features ($\mathbf{x}_1 = \mathbf{x}_2$), prove why the standard Lasso penalty can assign any arbitrary split of weights, whereas ElasticNet forces them to be equal.

Answer: Let us evaluate the behavior by examining the uniqueness of the solution under both penalties. Suppose we have a valid Lasso optimal allocation where the sum of the coefficients equals a total targeted

value: $\hat{\beta}_1 + \hat{\beta}_2 = c$.

Since $\mathbf{x}_1 = \mathbf{x}_2$, any alternative weight configuration $\tilde{\beta}_1 = \hat{\beta}_1 + \epsilon$ and $\tilde{\beta}_2 = \hat{\beta}_2 - \epsilon$ generates the exact same predicted value stream, leaving the Residual Sum of Squares (RSS) completely unchanged.

Let us analyze how this shift impacts the L_1 penalty term, assuming both coefficients remain positive:

$$\|\tilde{\boldsymbol{\beta}}\|_1 = |\hat{\beta}_1 + \epsilon| + |\hat{\beta}_2 - \epsilon| = \hat{\beta}_1 + \epsilon + \hat{\beta}_2 - \epsilon = \hat{\beta}_1 + \hat{\beta}_2 = \|\hat{\boldsymbol{\beta}}\|_1$$

Because the total loss remains constant for any value of ϵ, the Lasso optimization problem lacks strict convexity. This allows the algorithm to choose an arbitrary weight split based entirely on minor numerical round-off artifacts or the sequence of your feature columns.

Now, let us introduce the quadratic L_2 penalty component ($\lambda_2\|\boldsymbol{\beta}\|_2^2$) present in ElasticNet, and evaluate the same weight adjustment:

$$\|\tilde{\boldsymbol{\beta}}\|_2^2 = (\hat{\beta}_1 + \epsilon)^2 + (\hat{\beta}_2 - \epsilon)^2 = \hat{\beta}_1^2 + 2\epsilon\hat{\beta}_1 + \epsilon^2 + \hat{\beta}_2^2 - 2\epsilon\hat{\beta}_2 + \epsilon^2 = \|\hat{\boldsymbol{\beta}}\|_2^2 + 2\epsilon(\hat{\beta}_1 - \hat{\beta}_2) + 2\epsilon^2$$

To find the value of ϵ that minimizes this penalty, we differentiate with respect to ϵ and set the derivative to zero:

$$\frac{\partial}{\partial \epsilon}\left(\|\tilde{\boldsymbol{\beta}}\|_2^2\right) = 2(\hat{\beta}_1 - \hat{\beta}_2) + 4\epsilon = 0 \implies \epsilon^* = \frac{\hat{\beta}_2 - \hat{\beta}_1}{2}$$

Substituting this optimal ϵ^* back into our alternative weight expressions:

$$\tilde{\beta}_1 = \hat{\beta}_1 + \frac{\hat{\beta}_2 - \hat{\beta}_1}{2} = \frac{\hat{\beta}_1 + \hat{\beta}_2}{2}, \quad \tilde{\beta}_2 = \hat{\beta}_2 - \frac{\hat{\beta}_2 - \hat{\beta}_1}{2} = \frac{\hat{\beta}_1 + \hat{\beta}_2}{2} \implies \tilde{\beta}_1 = \tilde{\beta}_2$$

This result demonstrates that the L_2 component adds strict curvature to the loss surface. This quadratic penalty forces identical or highly correlated features to take equal weights, providing the numerical stability required for live trading applications.

Question 2: How does the "Irrepresentable Condition" limit Lasso's ability to select the correct underlying causal features in an equities asset book?

Answer: The Irrepresentable Condition is a structural requirement that determines whether Lasso can successfully identify the true causal signals while filtering out uninformative noise. Let the feature matrix be partitioned into two groups: $\mathbf{X}_{\text{active}}$ (the true causal signals) and $\mathbf{X}_{\text{noise}}$ (the non-causal features). The condition requires that:

$$\left|\mathbf{X}_{\text{noise}}^\top \mathbf{X}_{\text{active}} \left(\mathbf{X}_{\text{active}}^\top \mathbf{X}_{\text{active}}\right)^{-1} \text{sign}(\boldsymbol{\beta}_{\text{active}})\right| < \mathbf{1}$$

This matrix expression states that the regression coefficients obtained by projecting the noise features onto the active features, weighted by the signs of the true signals, must be strictly less than 1 across all coordinates.

If this condition is violated—which occurs frequently in equity markets where noise features (such as industry exchange-traded funds) correlate heavily with specific corporate alpha signals—Lasso cannot isolate the true causal drivers.

Instead, the regularizer will systematically select non-causal noise features, corrupting your feature selection mask and undermining the predictive power of the model when applied to out-of-sample data.